Sahar Levy

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Professional Summary

U.S. citizen and mechanical engineering student at Cal Poly Pomona with hands-on experience in CAD modeling, mechanism design, additive manufacturing, and shop fabrication. Contributor to multidisciplinary engineering teams spanning unmanned aircraft, rocket materials research, and award-winning competition design. Skilled at taking mechanical systems from concept through design, prototyping, and testing, with a strong interest in materials engineering. Recognized with the Gene Haas Foundation Scholarship for technical aptitude and leadership.

Education

California State Polytechnic University, Pomona
B.S., Mechanical Engineering. Fall 2025 - Present. GPA: 3.43. Expected Graduation 2029

Irvine Valley College
Design, Model Making and Rapid Prototyping Classes. Spring 2024 - Summer 2025

Woodbridge High School, Irvine
High School Diploma. GPA: 4.18. Graduated 2025

Experience

IAM3D Competition Team Member, 2025 - Present, *Cal Poly Pomona*
- Design and optimize land rover chassis structures in SolidWorks, balancing strength, weight, and mission requirements.
- Apply DFMA and DFAM principles to prototype and iterate on critical components using thermoplastics.
- Earned first place at the ASME eFX Dallas competition out of 20 national teams.

Liquid Rocket Lab Team Member, 2026 - Present, *Cal Poly Pomona*
- Verify mechanical properties of metal 3D-printed parts against published values after prior teams could not replicate them, intended for future launch vehicle and rocket engine use.
- Model test specimens in Fusion 360 and develop test procedures following ASTM standard E8.
- Run tensile tests on specimens and write twice-yearly reports to be delivered to sponsors.

Banshee UAV Team Member, 2025 - Present, *Cal Poly Pomona*
- Fabricate and assemble VTOL UAV components including XPS foam wing cutting and 3D printing.
- Coordinate across aerodynamics, avionics, design, flight testing, and robotics subteams to verify structural integrity and operational readiness.
- Team sponsored by Lockheed Martin, Southern California Edison, Air Force Research Laboratory, Robotis, and General Atomics.

Activities & Leadership

American Society of Mechanical Engineers (ASME), 2025 - Present, *Cal Poly Pomona*
Engage with the professional engineering community through industry speaker events and workshops.

SkillsUSA Executive Member, 2025, *Woodbridge High School*
Led and organized chapter events and competition preparations, and promoted STEM collaboration and career readiness.

COSMOS Program Participant, Summer 2024, *University of California, Irvine*
A highly selective four-week University of California program in collaboration with UCI faculty, culminating in a team project analyzing hybrid-series aircraft propulsion with a symposium presentation.

Scouts Counselor, 2020 - 2025, *Orange County Jewish Scouts*
Organized weekly programs and camps for youth groups, handling logistics and safety while mentoring younger members.

Awards & Honors

- 1st Place, ASME eFX Dallas Competition (2025)
- Cal Poly Pomona Dean's List (2025)
- Gene Haas Foundation Scholarship Recipient (2025)
- SkillsUSA Gold Chapter of Distinction (2025)
- SkillsUSA Automated Manufacturing Technology State Qualifier (2025)
- Presidential Volunteer Service Award (2023, 2024)

Skills (please view portfolio)

Technical: CAD Modeling (SolidWorks, Fusion 360), Technical Drawings, MATLAB, FDM and Metal Additive Manufacturing / 3D Printing, Structural Design, Rapid Prototyping, Manual Machining, Metal Casting, Shop Experience